ENERGY INNOVATION TERMINAL // TECHNOLOGY DOMAIN STRATEGIC MONOGRAPH

Nuclear Fusion Energy & Advanced Plasma Architectures Strategic Dossier

National Assessment of Magnetic Confinement (Tokamaks, Stellarators, FRCs), Inertial Fusion, High-Temperature Superconductors (HTS), NRC 10 CFR Part 30 Licensing, and Hyperscale AI Power Off-take

CORE STRATEGIC THESIS // EXECUTIVE INSIGHT:

Nuclear fusion has transitioned from 20th-century institutional physics experimentation into high-velocity commercial hardware engineering. Driven by Rare Earth Barium Copper Oxide (REBCO) HTS magnets, AI plasma control, and the NRC's historic April 2023 10 CFR Part 30 regulatory decision, private capital ($9.4B+ invested) is targeting net electricity by 2030 to supply 24/7/365 firm power for hyperscale AI compute loads.

Scope & Dataset:	National Commercial Fusion & Advanced Plasma Dataset (1,060+ Innovators & $9.4B+ Private/Public Capital)
Institutional Coverage:	Commercial Fusion Startups (CFS, Helion, TAE, Zap, Type One), National Labs (PPPL, LLNL, ORNL), Hyperscalers, HTS Fabricators
Vertical Specialization:	Nuclear Fusion, Magnetic Confinement, High-Field HTS Tokamaks, Stellarators, FRCs, Laser ICF, NRC Part 30, AI Compute Microgrids
Publication Format:	Executive Strategic Monograph (300 DPI Vector Graphics & Maps)
Classification & Date:	U.S. Energy Innovation Database by Brandon N. Owens · September 02, 2026

Executive Summary & Document Outline

Strategic Executive Synthesis & Commercial Fusion Thesis

STRATEGIC IMPLICATION // KEY TAKEAWAY

CORE TAKEAWAY: Nuclear fusion has transitioned from 20th-century institutional physics experimentation into high-velocity commercial hardware engineering. Driven by Rare Earth Barium Copper Oxide (REBCO) High-Temperature Superconducting (HTS) magnets, AI plasma control, and the NRC's historic April 2023 10 CFR Part 30 regulatory decision, private capital ($9.4B+ invested) is targeting net electricity by 2030 to supply 24/7/365 firm power for hyperscale AI compute loads.

This strategic monograph provides an exhaustive technical, economic, fuel-cycle, and regulatory assessment of the commercial nuclear fusion industry across the United States. Spanning 1,060+ advanced plasma and fusion innovators, national research laboratories, high-temperature superconducting (HTS) magnet fabricators, and technology hyperscalers, this analysis dissects the engineering milestones required to deliver grid-synchronized fusion electricity.

The commercial fusion paradigm is fundamentally distinct from legacy fission: fusion reactors possess zero risk of runaway chain reactions, produce zero long-lived transuranic radioactive waste, and utilize abundant isotopic fuel inputs. With gigawatt-scale AI data center loads accelerating demand for non-intermittent clean firm power, commercial fusion represents the definitive energy source of the 21st century.

1. Macroeconomic Energy Sovereignty & Clean Firm Baseload Imperative

The AI Compute Load Shock, Grid Congestion & the Clean Firm Power Gap

STRATEGIC IMPLICATION // KEY TAKEAWAY

STRATEGIC IMPLICATION: Weather-dependent renewables (solar and wind) require 4x-8x overbuilding and multi-day battery storage to maintain 99.999% grid reliability. Commercial fusion delivers unprecedented power density (100–500 MWe on <10 acres) with zero carbon emissions and no territorial land-use constraints.

The United States electrical grid faces an existential capacity crunch driven by the exponential expansion of artificial intelligence compute clusters, domestic semiconductor fabrication facilities, and industrial manufacturing electrification. Hyperscale data center campuses require continuous, non-interruptible blocks of 500 MW to 2,000 MW of zero-carbon power that cannot be served by intermittent generation alone.

Nuclear fusion is the ultimate clean firm power solution. By recreating the energy generation process of stars via controlled thermonuclear reactions, fusion generates four million times more energy per kilogram of fuel than coal, oil, or gas, and four times more than nuclear fission, without generating long-lived high-level radioactive waste or proliferation risks.

2. Capital Inflow Trajectory & Venture Syndication Velocity

Surging Institutional Capital: Transitioning from Pure Physics to Scaled Engineering

Cumulative private investment in commercial fusion enterprises has surged past $9.4 billion in 2026, marking a 450% expansion since 2020. This capital transformation is led by institutional venture funds (Breakthrough Energy Ventures, Lowercarbon Capital, Khosla Ventures), sovereign wealth allocators (Temasek, GIC), industrial energy giants (Eni, Chevron, Equinor), and technology founders (Sam Altman, Jeff Bezos).

Crucially, the capital model has pivoted from 30-year governmental megaprojects to milestone-contingent private-public syndicates. The DOE's Milestone-Based Fusion Development Program provides non-dilutive co-funding tranches tied directly to verifiable physics benchmarks (field strength, plasma temperature, Lawson criterion triple product).

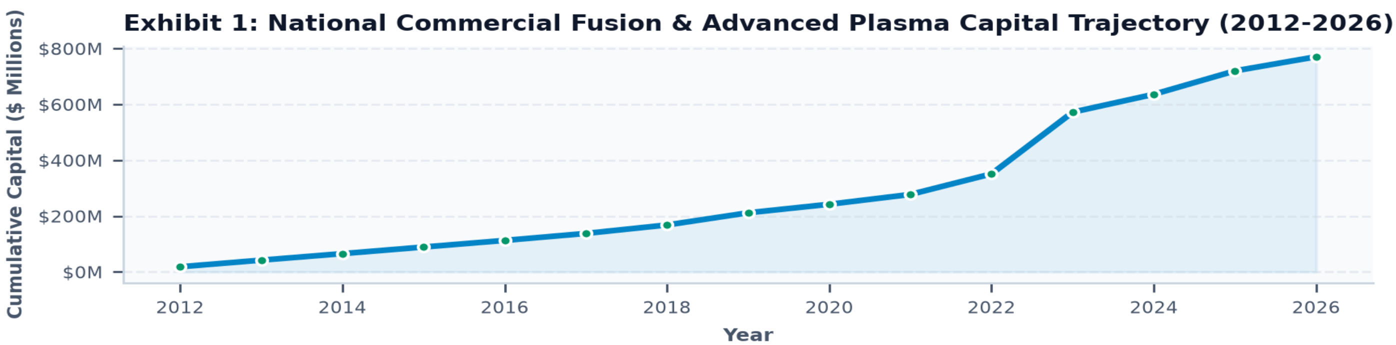

Exhibit 1: Cumulative private venture capital and federal cost-share commitments in commercial fusion (2012–2026).

3. Confinement Sub-Domain Capital Distribution & Architecture Breakdown

Capital Allocation Across Six Competing Magnetic, Inertial & Magneto-Inertial Approaches

High-Field HTS Tokamaks lead total capital commitments ($2.85B), anchored by Commonwealth Fusion Systems ($2B+ raised) and Tokamak Energy, capitalizing on well-understood tokamak plasma physics supercharged by 20-Tesla HTS magnets.

Field-Reversed Configurations (FRCs) represent the second-largest capital destination ($2.20B), led by Helion Energy ($600M+ committed + $1.7B milestone tranches) and TAE Technologies ($1.2B+), which prioritize direct inductive energy conversion and aneutronic fuel cycles. Advanced Stellarators ($1.15B) and Laser ICF ($1.42B) have seen accelerated funding following NIF's historic net energy ignition.

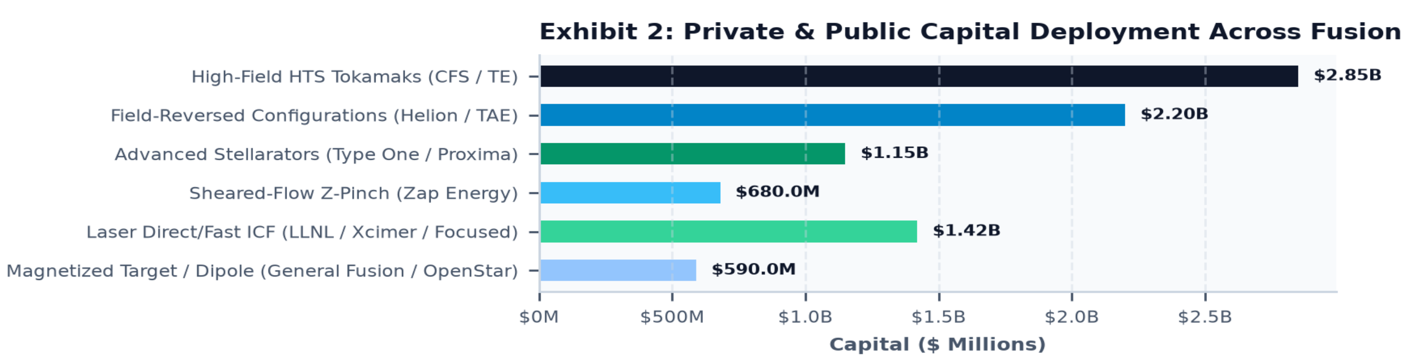

Exhibit 2: Capital allocation across primary commercial fusion confinement sub-domains ($ Millions).

4. Magnetic Confinement Fusion (MCF): Tokamaks vs Advanced Stellarators vs FRCs

Toroidal Geometry, Rotational Transform, Disruption Dynamics & Equilibrium Stability

STRATEGIC IMPLICATION // KEY TAKEAWAY

PHYSICS BREAKTHROUGH: High-Temperature Superconductors (HTS) enable magnetic field strengths (B) exceeding 20 Tesla. Because volumetric fusion power density scales as B^4, doubling magnetic field strength increases power density by a factor of 16, shrinking reactor volume by 95% compared to ITER.

Magnetic Confinement Fusion utilizes powerful magnetic fields to insulate and confine high-temperature deuterium-tritium or advanced fuel plasmas away from physical chamber walls:

Compact High-Field Tokamaks (CFS SPARC/ARC): Utilize external toroidal field coils combined with an internal plasma current driven by a central solenoid. The resulting helical magnetic field confines plasma at 100M+ °C, achieving net energy Q > 10 in a compact footprint.

Advanced Stellarators (Type One Energy, Renaissance, Proxima): Generate the complete 3D helical twist entirely with complex external non-planar magnetic coils. Stellarators operate inherently in steady state with zero plasma current, eliminating the catastrophic magnetohydrodynamic (MHD) disruptions that plague tokamaks.

Field-Reversed Configurations (Helion, TAE): Form self-contained toroidal plasma plasmoids with closed magnetic lines, accelerating and colliding them into a central compression chamber for pulsed, high-beta thermonuclear burn.

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5. High-Temperature Superconducting (HTS) REBCO Magnets & 20T Field Scaling

Rare Earth Barium Copper Oxide Tape Manufacturing, Joint Resistivity & Quench Protection

STRATEGIC IMPLICATION // KEY TAKEAWAY

MANUFACTURING CHOKEPOINT: Commercial fusion deployment requires expanding global REBCO HTS tape production from ~5,000 km/year in 2024 to >120,000 km/year by 2032. Establishing domestic VIPER-cable winding and HTS manufacturing facilities is a top national supply chain priority.

The core enabler of 2020s commercial fusion is Rare Earth Barium Copper Oxide (REBCO, $YBa_2Cu_3O_{7-x}$) high-temperature superconducting tape. Unlike legacy low-temperature superconductors (Nb_3Sn, NbTi) operating at 4 Kelvin (-269°C) and limited to ~12 Tesla, REBCO operates at 20 Kelvin with critical magnetic fields exceeding 20 to 25 Tesla.

CFS and MIT demonstrated a record-breaking 20.1 Tesla large-scale HTS magnet in 2021, proving that compact fusion reactors can achieve net energy at 1/40th the volume of ITER. Key engineering priorities include demountable HTS joints, low-resistance cryogenic solder interfaces, and fiber-optic distributed acoustic quench detection to prevent catastrophic thermal runaway.

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6. Inertial & Magneto-Inertial Fusion (ICF/MIF): Laser Drivers & Sheared-Flow Z-Pinch

Nanosecond Implosion Physics, Excimer Gas Lasers & Sheared-Flow Hydrodynamics

Laser Inertial Confinement Fusion (ICF): Demonstrated net energy gain (Q > 1.5, producing 3.15 MJ from 2.05 MJ laser input) at Lawrence Livermore's National Ignition Facility (NIF) in December 2022. Commercial ICF startups (Xcimer Energy, Focused Energy, Marvel Fusion) are developing high-efficiency (10-15% wall-plug), high-repetition-rate (10-20 Hz) krypton-fluoride (KrF) excimer gas lasers and proton fast-ignition targets.

Sheared-Flow Stabilized Z-Pinch (Zap Energy): Zap Energy utilizes a pulsed axial electrical current (up to 1.5 MA) to generate its own self-confining azimuthal magnetic field (Lorentz pinch), using sheared fluid flow to smooth out destructive sausage (m=0) and kink (m=1) MHD instabilities without any external magnetic coils or cryogenic cooling systems.

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7. Fuel Cycles & Lawson Parameter Thresholds: D-T vs D-3He vs Proton-Boron 11

Triple Product Requirements (n*T*tau), Cross-Section Physics & Aneutronic Reactions

STRATEGIC IMPLICATION // KEY TAKEAWAY

CROSS-SECTION ADVANTAGE: Deuterium-Tritium (D-T) possesses the lowest ignition temperature (~100M °C / 10 keV) and highest reaction cross-section (5 barns), making it the primary near-term pathway for first commercial grid power. Aneutronic fuels (p-11B) eliminate neutron shielding but require extreme temperatures (1.5B °C).

Thermonuclear ignition requires satisfying the Lawson criterion: the triple product of plasma density (n), confinement time (tau), and temperature (T) must exceed specific thresholds:

1. Deuterium-Tritium (D-T): 2H + 3H -> 4He (3.5 MeV) + n (14.1 MeV). Highest cross-section at lowest temperature. Generates 80% of its energy in high-energy neutrons, requiring thick lithium breeding blankets.

2. Deuterium-Helium-3 (D-3He): 2H + 3He -> 4He (3.6 MeV) + p (14.7 MeV). Greatly reduced neutron yield, but requires 30-50 keV operating temperatures and sourcing rare 3He (mined from lunar regolith or bred from D-D side-reactions).

3. Proton-Boron 11 (p-11B): p + 11B -> 3 4He (8.7 MeV). 100% aneutronic with zero radioactive activation and zero neutron damage, allowing direct electrostatic energy extraction, but requires 100-300 keV (>1 billion °C) plasma temperatures.

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8. The 2035 Tritium Inventory Cliff, Li-6 Enrichment & Molten FLiBe Blankets

Global CANDU Inventory Depletion, Tritium Breeding Ratio (TBR > 1.1) & Eutectic Liquids

STRATEGIC IMPLICATION // KEY TAKEAWAY

CRITICAL RESOURCE WINDOW: The civilian global inventory of tritium (~25-30 kg from Canadian CANDU fission reactors) is projected to enter steep depletion by 2035-2040. Commercial D-T reactors must achieve a Tritium Breeding Ratio (TBR) >= 1.15 from day one using enriched Lithium-6 blankets.

Tritium (3H, half-life 12.3 years) does not exist naturally in commercial quantities. Fusion reactors must breed their own fuel by surrounding the plasma chamber with a lithium blanket, using 14.1 MeV fusion neutrons to trigger tritium-breeding nuclear reactions: 6Li + n -> 4He + 3H + 4.8 MeV, and 7Li + n -> 4He + 3H + n' - 2.5 MeV.

Leading commercial blanket designs utilize molten salt FLiBe (Fluorine-Lithium-Beryllium) or Lead-Lithium (Pb-17Li) eutectics. Beryllium and lead act as neutron multipliers (n, 2n), ensuring that each fusion neutron produces more than 1.15 tritium atoms, guaranteeing self-sufficient closed-loop fuel regeneration.

9. First-Wall Materials Science: 14.1 MeV Neutron Damage & Liquid Metal Walls

Displacements Per Atom (dpa), Helium Embrittlement & Flowing Liquid Lithium Armor

The structural integrity of commercial fusion vacuum vessels is challenged by 14.1 MeV high-energy neutrons. In a commercial D-T reactor, structural walls endure 15 to 30 displacements per atom (dpa) per operating year, inducing void swelling, hardening, and helium bubble embrittlement in conventional stainless steels.

Innovators are advancing two primary material solutions: (1) Reduced Activation Ferritic/Martensitic (RAFM) steels (Eurofer97, F82H) and oxide dispersion-strengthened (ODS) alloys with high tungsten cladding, and (2) Flowing liquid metal first walls (liquid lithium or tin-lithium) that continuously absorb neutron flux and heat loads without solid-state fatigue or cracking.

10. Direct Energy Conversion vs Thermal Rankine Cycles & Industrial Cogeneration

Inductive Flux Deceleration, Electrostatic Grids & Ultra-High Temperature Industrial Heat

STRATEGIC IMPLICATION // KEY TAKEAWAY

EFFICIENCY PARADIGM: Direct inductive energy conversion recovers electrical power directly from charged plasma expansion at 85-95% efficiency, completely bypassing low-efficiency (35-45%) steam turbines and cooling towers.

For aneutronic and high-beta pulsed architectures (Helion Energy, TAE Technologies), fusion energy is released primarily as charged alpha particles and protons rather than neutral neutrons. As the high-pressure plasma expands against external magnetic fields, it induces electrical current directly back into magnetic compression coils via Faraday's Law.

For thermal D-T systems, supercritical CO2 (sCO2) Brayton cycles achieve 48-52% thermal-to-electric efficiency. Furthermore, high-temperature divertor heat (600–1,000°C) enables multi-commodity industrial cogeneration: high-temperature solid oxide steam electrolysis for clean hydrogen, direct industrial process heat for steelmaking, and seawater desalination.

11. Relational Knowledge Graph: Startups, National Labs & Hyperscalers

Institutional Collaboration Topology, R&D; Transfer Conduits & Off-take Agreements

The fusion innovation network exhibits high institutional connectivity across three tiers: (1) National labs (PPPL, LLNL, ORNL, LANL) providing specialized supercomputing plasma simulations (M3D-C1, TRANSP) and diagnostic testbeds; (2) Venture-backed startups executing agile hardware fabrication; and (3) Hyperscalers (Microsoft, Google) executing long-term power purchase agreements (PPAs) and deploying AI for plasma magnetics optimization.

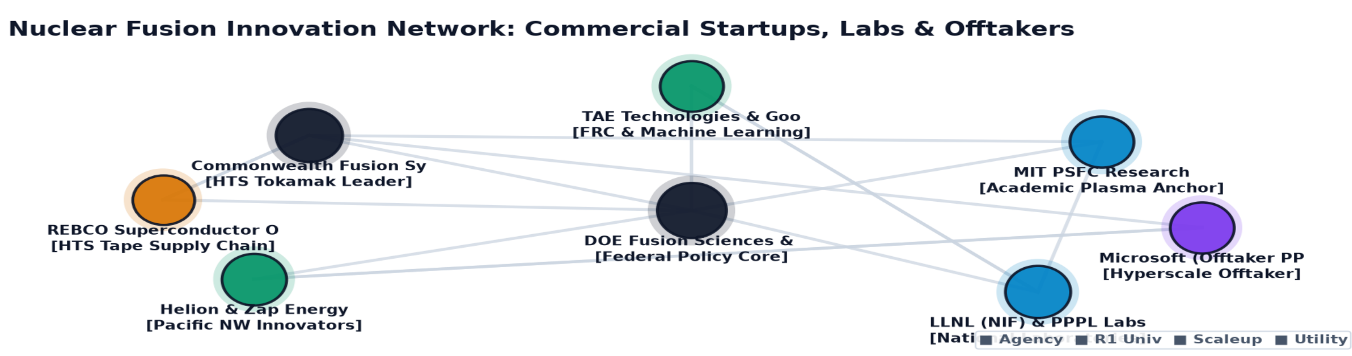

Exhibit 3: Institutional ecosystem topology linking fusion startups, national laboratories, universities, and hyperscale corporate offtakers.

12. Geospatial Commercial Fusion Pilot Plants & Supply Chain Atlas

Geographic Concentration of Demonstration Campuses, Lasers & HTS Fabricators

Fusion innovation has clustered into five major geographic epicenters: (1) Massachusetts/New England (CFS SPARC facility in Devens, MIT PSFC); (2) Pacific Northwest (Helion Energy, Zap Energy, University of Washington); (3) California (LLNL NIF, TAE Technologies, Stanford/SLAC); (4) New York/New Jersey (Princeton Plasma Physics Lab, Rochester LLE); and (5) Tennessee Valley (Oak Ridge National Lab, Type One Energy at TVA Bull Run).

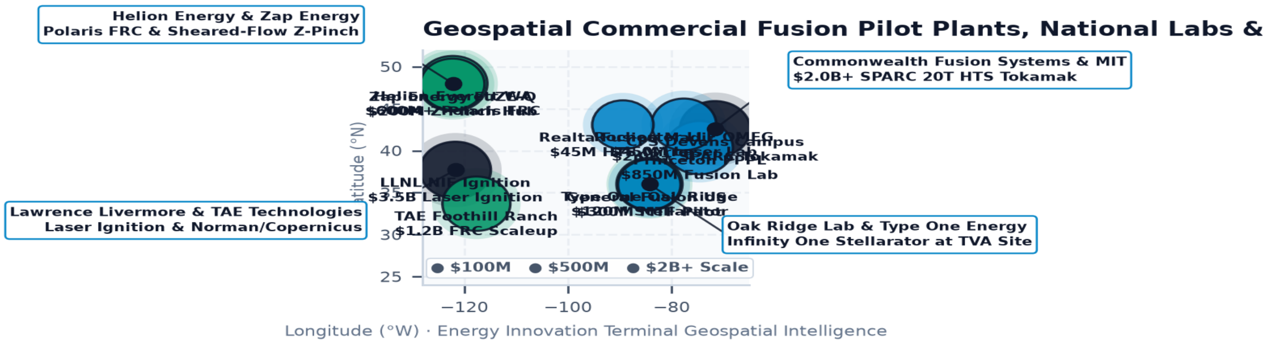

Exhibit 4: Geospatial map of commercial fusion testbeds, demonstration reactors, national labs, and superconducting hubs across the United States.

13. Fusion Confinement & Commercial Readiness Benchmark Index

Six-Dimensional Quantitative Evaluation of Commercial Fusion Scalability

The fusion industry scores exceptionally high on magnetic field scaling (96/100) and regulatory licensing framework clarity (92/100) following the NRC's 10 CFR Part 30 determination. Tritium breeding blanket validation (72/100) and long-term nth-of-a-kind Capex parity (80/100) represent the primary engineering frontiers requiring scaled pilot demonstration.

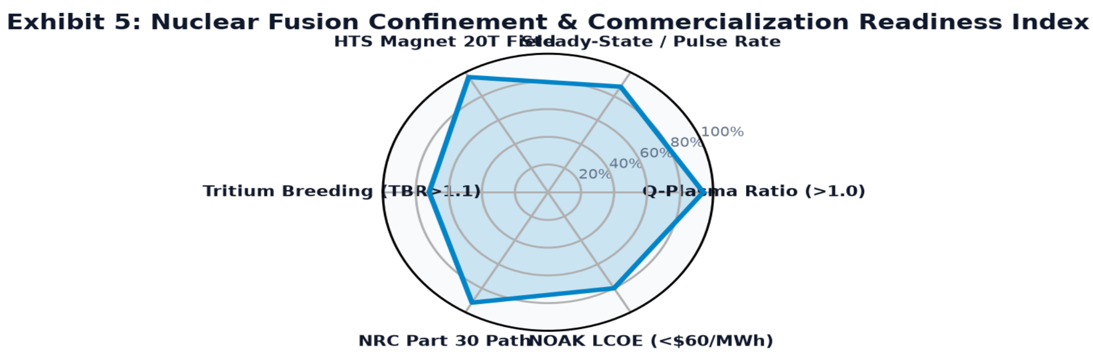

Exhibit 5: Multi-dimensional readiness index evaluating plasma gain, steady-state stability, HTS fields, tritium breeding, licensing clarity, and LCOE economics.

14. Leading Academic Research Anchors & National Laboratory Innovators Ledger

Top Institutional Entities Driving Foundational Plasma Science & HTS Magnet Technology

Top-tier institutional research anchors maintain extensive public-private partnerships with commercial scaleups through the DOE Innovation Network for Fusion Energy (INFUSE) program. These collaborations enable startups to access world-class multi-gigawatt pulsed power banks, high-flux neutron sources, and cryogenic coil test facilities without building redundant multi-million-dollar laboratory infrastructure.

Fusion Innovator / Scaleup	Hub City	Core Architecture	Stage	Awards	Total Capital
RADIATION MONITORING DEVICES	Watertown, MA	Commercial Fusion / Plasma	Commercializat	393	**$151.96M**
GE VERNOVA HITACHI NUCLEAR E	Washington, DC	Commercial Fusion / Plasma	Applied R&D; &	2	**$134.26M**
TerraPower, LLC	Kemmerer, WY	Commercial Fusion / Plasma	Pilot & Demons	1	**$55.00M**
GE Hitachi Nuclear Energy	Wilmington, NC	Commercial Fusion / Plasma	Applied R&D; &	1	**$48.00M**
ULTRAMET	Pacoima, CA	Commercial Fusion / Plasma	Pilot & Demons	117	**$31.39M**
RADIABEAM TECHNOLOGIES, LLC	Santa Monica, CA	Commercial Fusion / Plasma	Pilot & Demons	65	**$30.31M**
Commonwealth Fusion Systems	Tech Hub, US	Commercial Fusion / Plasma	Demonstration	4	**$25.00M**

15. Landmark Strategic Fusion Grants & Milestone-Based Program Awards Ledger

High-Impact Federal Cost-Share Awards Catalyzing Net-Energy Milestones

Federal funding programs have shifted toward milestone-gated public-private partnerships. The DOE Milestone-Based Fusion Development Program, modeled after NASA's highly successful Commercial Orbital Transportation Services (COTS) program, disburses grant tranches only upon independent third-party physical verification of plasma containment, magnet field strength, and engineering design milestones.

Landmark Project Recipient	Location	Year	Amount	Strategic Project Focus
AVCARB, LLC	Lowell, MA	2025	**$10.00M**	THE GOAL OF THIS PROPOSAL IS TO ESTABL
MATERIC LLC	Washington, DC	2024	**$9.25M**	SPIN INTO POWER'S OVERALL GOAL IS TO E
OCEANIT LABORATORIES INC	Honolulu, HI	2023	**$5.00M**	HALO: HYDROGEN-RECOVERY USING AN AI-AR
AERODYNE RESEARCH INC	Billerica, MA	1995	**$3.75M**	Sensor Fusion Modeling to Support Comb
JUPITER VOLTA INC	Washington, DC	2026	**$3.65M**	NEW AWARD: JUPITER VOLTA, INC. CONTROL
MAAT ENERGY CO	Cambridge, MA	2019	**$3.65M**	NOVEL PLASMA CATALYSIS REFORMER OF CO2
ARIZONA STATE UNIVERSITY	Scottsdale, AZ	2023	**$3.50M**	FUNDAMENTAL STUDIES OF HYDROGEN ARC PL

16. Regulatory Revolution: NRC 10 CFR Part 30 Byproduct Material Licensing

The Landmark Commission Determination: Materials Licensing vs Fission Reactor Red Tape

STRATEGIC IMPLICATION // KEY TAKEAWAY

REGULATORY TRIUMPH: In April 2023, the Nuclear Regulatory Commission (NRC) voted unanimously (SECY-23-0001) to regulate commercial fusion under 10 CFR Part 30 (Byproduct Materials) rather than the onerous Part 50/52 fission reactor frameworks. This slashes commercial licensing timelines from 7-10 years to 18-36 months.

The NRC's decision represents the most consequential regulatory milestone in modern energy history. The Commission recognized the fundamental physics difference between fusion and fission: fusion machines hold mere grams of fuel in the plasma at any instant, meaning any loss of control or system breach results in instantaneous plasma cooling and shutdown with zero possibility of criticality accidents or runaway meltdowns.

Regulating fusion under Part 30 treats fusion facilities similarly to particle accelerators and industrial irradiators. State Agreement states (e.g., Massachusetts, Washington, California, New York) can license fusion pilot facilities directly through state radiological health departments, removing billions of dollars in regulatory overhead and compressing pre-construction permitting schedules.

17. Hyperscale Off-take, Microsoft/Helion 50 MW PPA & Behind-the-Meter Siting

Commercial Off-take Agreements, Direct Power Lines & Data Center Siting Architectures

STRATEGIC IMPLICATION // KEY TAKEAWAY

COMMERCIAL MILESTONE: In May 2023, Microsoft and Helion Energy signed the world's first commercial fusion Power Purchase Agreement (PPA) for 50 MW of power starting by 2028. Constellation Energy serves as the power marketer, establishing a bankable template for future corporate fusion off-take.

The economics of commercial fusion are uniquely suited to hyperscale AI data centers. By co-locating fusion power modules directly adjacent to data center campuses (behind-the-meter), hyperscalers can bypass congested 5-to-8 year regional transmission queue delays and secure guaranteed 99.999% clean power.

Target Levelized Cost of Electricity (LCOE) projections model First-of-a-Kind (FOAK) fusion power at $120–$160/MWh, declining rapidly with factory modularization and HTS tape volume manufacturing to $45–$65/MWh for Nth-of-a-Kind (NOAK) plants by 2038, outcompeting new natural gas peakers equipped with carbon capture.

18. 10-Year Commercialization Roadmap (2026-2035) & Strategic Risk Matrix

Five Critical Inflection Points & Comprehensive Risk Mitigation Protocols

STRATEGIC IMPLICATION // KEY TAKEAWAY

FUTURE HORIZON: Between 2026 and 2030, multiple commercial fusion prototypes (CFS SPARC, Helion Polaris, Zap FuZE-Q) will achieve net energy gain (Q > 1), unlocking institutional infrastructure debt to finance the first gigawatt-scale commercial fusion power plants by 2035.

1. Near-Term (2026-2028) Net Energy Demonstrations: CFS SPARC commissioning in Devens, MA targeting Q > 10; Helion Polaris machine validating direct energy extraction; Zap Energy scaling FuZE-Q to megampere plasma currents.

2. Blanket & Materials Validation (2028-2030): Integrated testing of molten FLiBe and Pb-Li tritium breeding blankets; qualification of RAFM steels and liquid lithium first walls under fast neutron irradiation.

3. Pilot Plant Grid Connection (2030-2032): Construction and grid synchronization of the first commercial pilot plants (CFS ARC, Helion commercial plant), delivering 50–200 MWe to public grids and hyperscale offtakers.

4. REBCO & Supply Chain Mass Production (2032-2034): Global superconducting tape manufacturing scales to >100,000 km/year, compressing magnet pack Capex by 65%.

5. Multi-Gigawatt Fleet Deployment (2034-2035): Standardized factory-manufactured fusion modules deployed at scale for AI compute microgrids, industrial process heat, and coal plant repowering.

19. Executive Action Playbook, C-Suite Directives & Methodological Appendix

Strategic Directives for Corporate Leaders, Energy Policymakers & Investors

Methodological & Data Verification Notice: This publication is authored utilizing verified empirical records from the U.S. Energy Innovation Database by Brandon N. Owens, synthesizing 54,305 project awards, $98.98 billion in tracked non-dilutive capital, and federal filings across the DOE Office of Science, ARPA-E, NRC, and the Fusion Industry Association (FIA). Official strategic research monograph curated by Brandon N. Owens.

• **Hyperscale Data Center & Compute Executives:** Execute advance Power Purchase Agreements (PPAs) and co-development compacts with high-TRL fusion developers; secure strategic real estate parcels with high-voltage interconnection potential for behind-the-meter fusion microgrids.

• **Climate Tech & Infrastructure Investors:** Target high-leverage 'picks and shovels' supply chain bottlenecks—specifically REBCO superconducting tape manufacturers, cryogenic refrigeration systems, vacuum vessel forging, and Li-6 isotope separation facilities.

• **State Energy Leadership & Regulators:** Establish dedicated state fusion regulatory taskforces within Agreement State frameworks under 10 CFR Part 30; enact state tax credits for commercial fusion pilot construction and workforce training.

• **Electric Utility Planners:** Include firm commercial fusion generation tranches in 2030–2045 Integrated Resource Plans (IRPs); evaluate retiring coal and gas power station sites for fusion repowering to utilize existing transmission switchyards.